

Problem 7.1

Solution:**(a) Even-Length Loops ($\ell \geq 4$):**

Suppose an optimal solution contains an even-length loop $C = \langle v_1, v_2, \dots, v_{2k}, v_1 \rangle$ where $k \geq 2$. Since the vertices are distinct, we can decompose C into k disjoint 2-loops:

$$\langle v_1, v_2, v_1 \rangle, \langle v_3, v_4, v_3 \rangle, \dots, \langle v_{2k-1}, v_{2k}, v_{2k-1} \rangle.$$

The original loop has $2k$ edges, and the decomposed loops also contribute exactly $2k$ edges. Thus, replacing C with these 2-loops preserves optimality while eliminating even-length loops ($\ell \geq 4$).

Paths:

Assume an optimal solution contains a path $P = \langle v_1, v_2, \dots, v_{k+1} \rangle$ with k edges.

- If v_1 and v_{k+1} are adjacent, we can form a loop of length $k+1$, increasing the total edge count from k to $k+1$. This contradicts the optimality of the original solution, so such a path cannot exist in an optimal solution.
- If v_1 and v_{k+1} are not adjacent, consider the number of vertices $k+1$:
 - If $k+1$ is even, we can pair the vertices into $\frac{k+1}{2}$ 2-loops, contributing $k+1$ edges, which is better than the original k edges thus cannot exist.
 - If $k+1$ is odd, we can leave v_{k+1} unused and construct $\frac{k}{2}$ 2-loops, which does not decrease total length k and preserves optimality.

Thus, even-length loops and paths can always be replaced by 2-loops without decreasing the total length, so they do not appear in an optimal solution.

- (b)** Since the vertices of a bipartite graph can be partitioned into two independent sets A and B , every edge connects A and B . If an odd-length cycle existed, its start and end vertices would have to belong to the same set, which is not allowed in bipartite graph. Therefore, all cycles in a bipartite graph must have even lengths. According to part (a), we can conclude that the optimal solution consists solely of vertex-disjoint 2-loops.

Each 2-loop corresponds to a matching edge, contributing a length of 2. A collection of vertex-disjoint loops corresponds to a matching M , where the number of edges is $|M|$, and the total length is $2|M|$. Thus, the maximum matching k^* yields a total length of $2k^*$, which is the theoretical maximum.

Assume there exists a solution with a total length $L > 2k^*$. Then, the corresponding matching size would be $|M| = L/2 > k^*$, contradicting the fact that k^* is the maximum matching size. Therefore, the optimal total length must be $2k^*$.

- (c) Construct the Bipartite Graph G' :**

Split each vertex $v \in V$ into two copies v^L and v^R , placing them in separate partitions A and B of G' so that $A = \{v^L \mid v \in V\}$ and $B = \{v^R \mid v \in V\}$.

For each edge $(u, v) \in E$, add two edges in G' : (u^L, v^R) and (v^L, u^R) .

An odd-length $2k + 1$ cycle in G

$$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow \cdots \rightarrow v_{2k+1} \rightarrow v_1$$

forms a matching of size $2k + 1$ in G' :

$$v_1^L v_2^R, v_2^L v_3^R, v_3^L v_4^R, \dots, v_{2k+1}^L v_1^R.$$

A 2-loop $\langle u, v, u \rangle$ corresponds to selecting both $u^L v^R$ and $v^L u^R$ in G' .

Proof of Correctness:

i. From G to G' :

Every vertex-disjoint collection of loops in G (length-2 or odd) maps to a matching in G'

For a loop (length-2 or odd), match points alternately in G' , as stated above.

The total length in G is the size of the matching in G' .

ii. From G' to G :

Every matching in G' maps back to a vertex-disjoint collection of loops in G :

A matching of alternating edges

$$v_1^L v_2^R, v_2^L v_3^R, v_3^L v_4^R, \dots, v_n^L v_1^R$$

corresponds to a loop

$$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow \cdots \rightarrow v_n \rightarrow v_1$$

in G . From part (a), this is equivalent to a collection of vertex-disjoint 2-loops and odd-length loops.

A matching

$$v_1^L v_2^R, v_2^L v_3^R, v_3^L v_4^R, \dots, v_{n-1}^L v_n^R$$

corresponds to a path

$$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow \cdots \rightarrow v_n$$

in G . If such a matching exists in the maximum matching of G' , it implies that the vertices v_n^L and v_1^R are unmatched. However, by the construction of G' , there must exist an edge $v_n^L v_1^R$, which could have been matched, resulting in a loop in G . Hence, path cannot exist in G' 's maximum matching.

The property of matching guarantees that no vertex is used twice thus G is vertex-disjoint.

Therefore, the maximum matching of G' yields the maximum total length of G . G' has $2n$ vertices and $2m$ edges, which can be solved in $O(mn)$ time by the Hungarian method or $O(m\sqrt{n})$ time by Hopcroft and Karp's algorithm.